

MACS+ - Membrane Action Of Composite Structures In Case Of Fire

R E P O R T



1. Project properties

Project name:	Atlantic Park Phase 2 Unit 6	Prepared by:	KevinKurniawan
Client name:		Company name:	Fire Dynamics Group Limited
Client company:	Winvic Construction Limited	Date:	20 August 2025, 13:02
Job number:	0563	Checked by:	
Comments:	First and Second Floors Slab		

2. General arrangement

- Spans
 - Span 1: 7.1 m
 - Span 2: 9 m
- Unprotected beams
 - Number of internal unprotected beams: 2

3. Deck details

- Deck properties
 - Deck: Multideck 60 Type: Trapezoidal
 - Depth: 60 mm Top flange: 142 mm
 - Pitch: 323 mm Bottom flange: 119 mm
 - Stiffener height: 9 mm

4. Slab details

- Concrete
 - Concrete type: Normal Slab depth: 150 mm
 - Cylinder compressive strength of concrete (f_{ck}): 30 N/mm²
- Mesh
 - Mesh type: A193
 - Longitudinal mesh area: 193 mm²/m Bar size: 7 mm
 - Transverse mesh area: 193 mm²/m Bar size: 7 mm
 - Average mesh axis distance: 40 mm Mesh yield stress: 500 N/mm²

5. Beams details

- Unprotected beams
 - Beam type: Solid beam
 - Section family: UK sections Steel grade: S355
 - Degree of shear connection: 80 %
 - Section size: UB 356x127x33
 - Details: $h = 349$ mm, $b = 125.4$ mm, $t_w = 6$ mm, $t_f = 8.5$ mm
- Side A perimeter beam
 - Beam type: Solid beam
 - Section family: UK sections Steel grade: S355
 - Section size: UB 356x127x33
 - Details: $h = 349$ mm, $b = 125.4$ mm, $t_w = 6$ mm, $t_f = 8.5$ mm
 - Beam location: Internal beam Construction type: Composite
 - Degree of shear connection: 80 %

- Side B perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

- Side C perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, $t_w = 6$ mm, $t_f = 8.5$ mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

- Side D perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, $t_w = 7.6$ mm, $t_f = 10.9$ mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

6. Loading details

- Normal (Cold)

Leading variable action:	2.5 kN/m ²
Accompanying variable action:	0 kN/m ²
Dead load including beam, excluding slab:	2 kN/m ²
Calculated slab weight including mesh:	2.83 kN/m ²

- Fire (Hot)

Combination factor for permanent action:	1.0
Combination factor for leading variable action:	1
Combination factor for other variable action:	1

7. Fire & Analysis

- Parametric temperature-time curve

Compartment length:	53.2 m	Window height:	3.8 m
Compartment width:	7.2 m	Window length:	53.2 m
Compartment height:	3.8 m	Glazing breakage:	93.683 %
Fire load:	440.68 MJ/m ²	Wall lining (B) factor:	1700 J/m ² s ^{1/2} K
Growth rate:	Medium	Combustion factor:	0.8

8. Summary output

- Default mesh direction

Maximum unity factor:	0.36	Floor slab adequate
Factored load in fire:	7.33 kN/m ²	
Fire curve:	Parametric temperature-time curve	

9. Default mesh direction

- Tabular results

Longitudinal mesh area:	193 mm ² /m	Bar size:	7 mm
Transverse mesh area:	193 mm ² /m	Bar size:	7 mm

Factored load in fire: 7.33 kN/m²

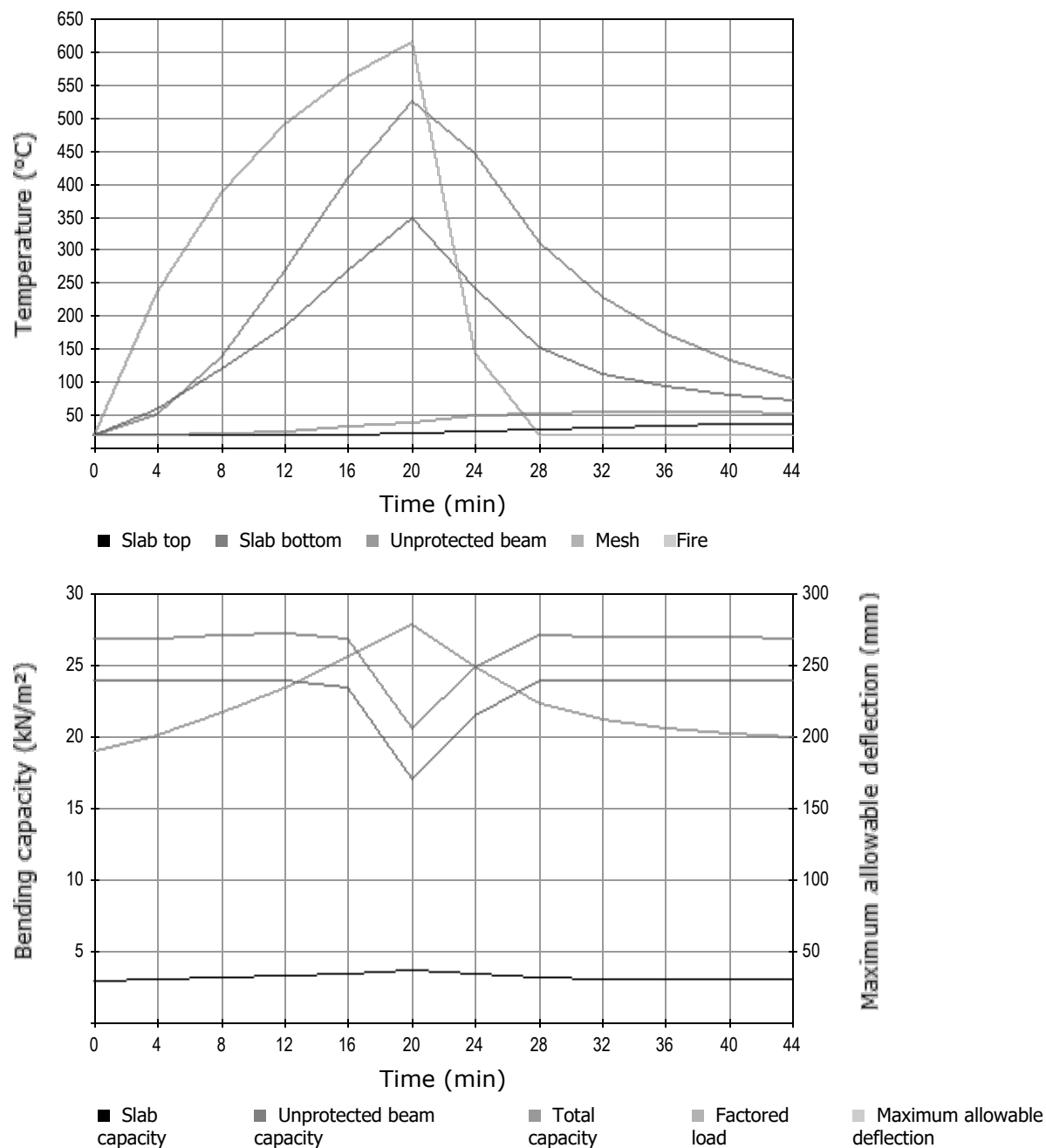
Time	Beam	Mesh	Slab top	Slab bottom	Beam capacity	Maximum allowable deflection	Slab yield	Enhancement	Slab capacity	Total capacity	Unity factor
mins	°C	°C	°C	°C	kN/m ²	mm	kN/m ²		kN/m ²	kN/m ²	
0	20	20	20	20	23.88	190	1.41	2.07	2.92	26.80	0.27
4	52	20	20	60	23.88	201	1.41	2.13	3.00	26.88	0.27
8	141	23	20	120	23.88	217	1.41	2.22	3.14	27.01	0.27
12	270	28	20	185	23.88	234	1.41	2.32	3.28	27.15	0.27
16	409	33	21	268	23.41	256	1.41	2.45	3.46	26.87	0.27
20	525	41	23	350	17.02	278	1.41	2.58	3.63	20.65	0.36
24	447	50	25	244	21.50	249	1.41	2.41	3.39	24.90	0.29
28	312	54	28	152	23.88	223	1.41	2.26	3.19	27.07	0.27
32	229	55	31	114	23.88	212	1.41	2.20	3.10	26.98	0.27
36	173	55	34	94	23.88	206	1.41	2.16	3.05	26.93	0.27
40	135	55	36	82	23.88	202	1.41	2.14	3.02	26.90	0.27
44	106	54	38	73	23.88	200	1.41	2.12	2.99	26.87	0.27

Maximum unity factor: 0.36 **Floor slab adequate**

• Perimeter beam check

Side A	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 356x127x33	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.71	
	Critical temperature:	544 °C	
Side B	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.71	
	Critical temperature:	541 °C	
Side C	Beam type:	Solid beam	Composite Internal beam
	Section size:	UB 356x127x33	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.71	
	Critical temperature:	544 °C	
Side D	Beam type:	Solid beam	Composite Edge beam
	Section size:	UB 457x152x52	
	Required moment resistance in fire situation:	0 kNm	
	Line load in fire situation:	0 kN/m	
	Shear connection:	80 %	
	Degree of utilization:	0.71	
	Critical temperature:	541 °C	

10. Graphical output



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